ISIC-2024 SLICE-3D: Skin-Cancer Detection on a Quality-vs-Cost Frontier

Single-dataset, no-external-data, no-synthetic study, reported on a pAUC-versus-cost Pareto frontier.

Group **Pakistan.AI** · Final Project, Neural Networks · National Yunlin University of Science and Technology

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CLINICAL TASK

Melanoma Triage From Total-Body Photography

- Melanoma is highly survivable when excised early; mortality rises sharply once it metastasises.
- The clinical bottleneck is **triage**: ranking which lesions warrant dermatologist review.
- ISIC-2024 frames this as a ranking task over lesions imaged by **3D total-body photography** (TBP), not dermoscopy.
- Each lesion is a small square **crop** plus per-lesion tabular metadata.

> Objective: rank a patient's lesions so the **highest-risk are reviewed first**, using only signals a routine total-body scan already captures.

SLICE-3D: 401,059 Lesion Crops

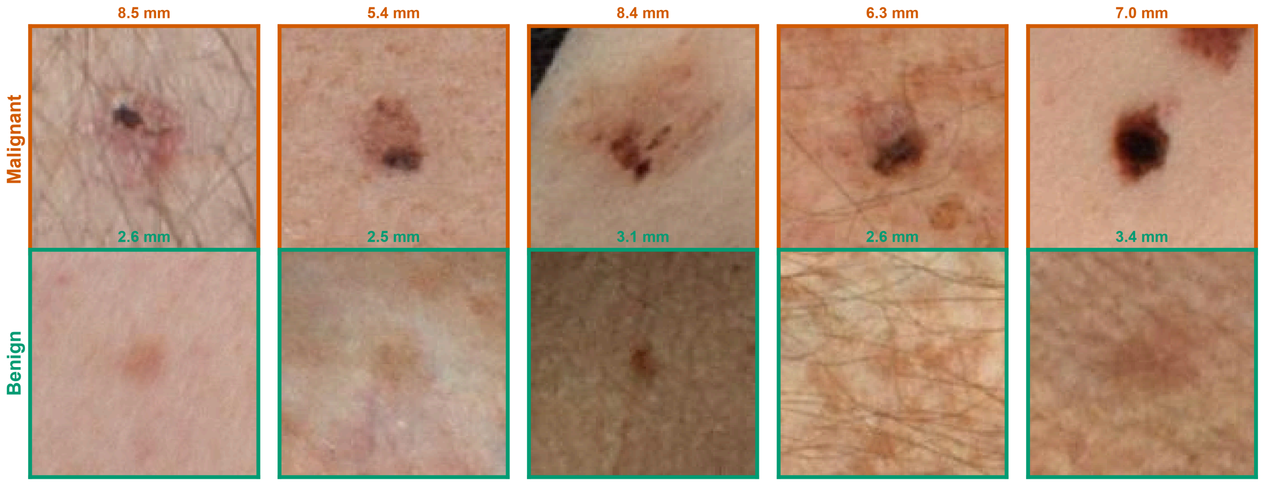
401,059
lesion crops

393
malignant

0.098%
prevalence

1,042
patients

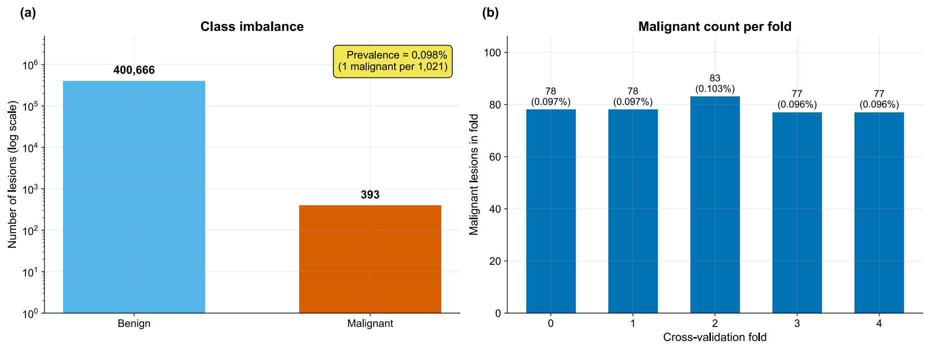
Representative lesion crops by class



Malignant lesions trend larger, more colour-heterogeneous and more border-irregular; benign nevi are smaller and more uniform.
Each crop is annotated with its clinical longest diameter.

- Square TBP tiles, 61–239 px (median 131 px).
- 393 malignant against **400,666** benign.
- One cancer per **~1,021** crops.
- Head/neck site carries ~7× the baseline malignant rate.

Extreme Imbalance, Scored on the High-Sensitivity Tail



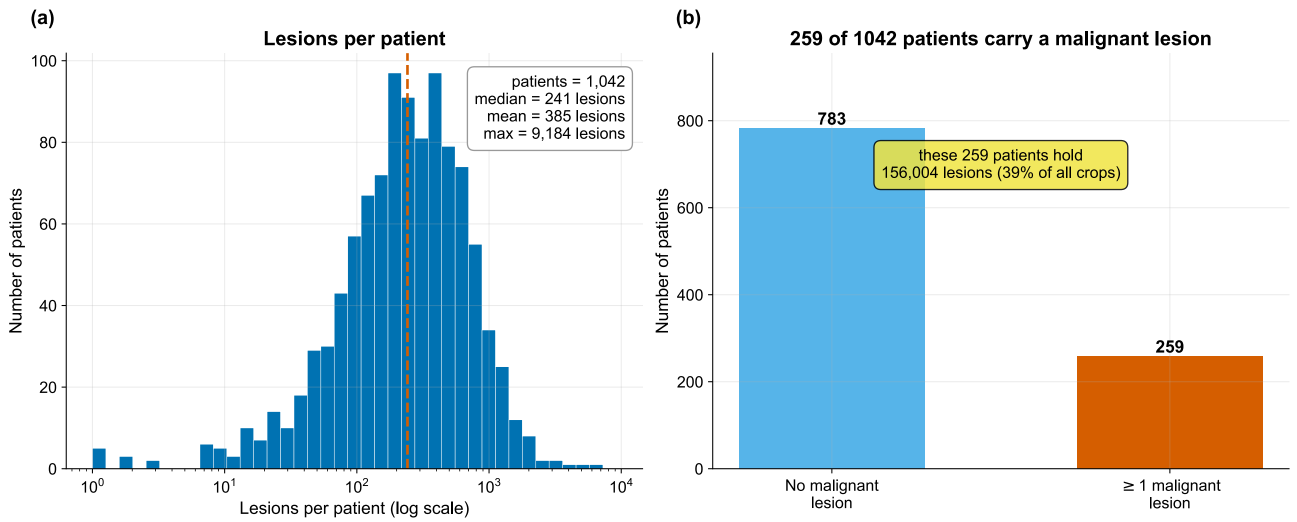
Class counts (log scale) and the 77–83 malignant cases per fold.

- At 0.098% prevalence, plain AUC and accuracy reward the wrong region of the curve.
- Scoring is restricted to the **high-sensitivity tail**: TPR from 0.80 to 1.0.
- Range **[0, 0.20]**: random ≈ 0.02, perfect = 0.20.

$$\mathrm{pAUC}_{80} = \int_{0.80}^{1.0} (1 - \mathrm{FPR})\, d(\mathrm{TPR})$$

Official ISIC-2024 metric. Every number here is OOF pAUC@80%TPR on the same frozen folds.

Patient-Grouped, Target-Stratified 5-Fold



Lesions cluster by patient; folds split on patients, not crops.

- **StratifiedGroupKFold**: grouped by patient, stratified on the label.
- **No patient straddles two folds.**
- One frozen **folds.parquet**; every model reads the identical split.
- Each fold holds 77–83 positives, so none is starved of signal.

WHY IT MATTERS

If a patient's moles appear in both train and validation, the model memorises the person, not the disease — the source of the competition's public-to-private shake-ups.

APPROACH

Two Experts, A Trivial Stack

Data · 401k crops + metadata → Frozen **patient-grouped** 5-fold split

Tabular | Intrinsic + **patient-relative** features → Bagged **LightGBM + CatBoost**

Image | **convnextv2_nano** @224, undersample → OOF **prob + embedding**

Tabular OOF + Image OOF → **Rank-average** → **Submission**

> Tabular expert carries most of the score; the image expert contributes a complementary signal; the combiner is a zero-parameter rank average. No external or synthetic data.

- Intrinsic features: lesion geometry, colour, size ratios.
- **Patient-relative** deviations encode the ugly-duckling sign — a lesion that departs from its patient's own moles.
- 392 of 393 cancers share a patient with benign moles, so within-patient deviation is computable for nearly every positive.
- Bagged **LightGBM** ensembled with **CatBoost**; early-stop on pAUC@80%TPR.

> Patient-relative features supply 6 of the top 7 by gain and account for ~65% of total feature importance.

Table 1. Tabular LightGBM progression (OOF, leak-verified).

Configuration	OOF pAUC@80%TPR
Step 0 — broken: is_unbalance + AUC early-stop	0.09941
Step 1 — fixed early-stopping / objective	0.11826
Step 2 — wide patient-relative features	0.14420
Step 3 — bagged + cross feats + CatBoost	**0.16890**

- Small pretrained backbones; ImageNet weights carry no skin-cancer labels, preserving the no-external-data claim.
- Per-epoch **negative undersampling** against 0.098% prevalence.
- BCE with label smoothing, classical **transV2 augmentation**, light **EMA**, and mixup.
- Each backbone emits an OOF malignancy probability plus an embedding, stacked into the GBDT.

> Resolution and light regularisation on a small backbone is the sweet spot: convnextv2_nano gains **+0.005** from 128 to 224 px; stronger EMA over-smooths and hurts.

Image backbones (OOF pAUC@80%TPR).

Backbone	pAUC
ConvNeXtV2-nano @224	**0.15821**
ConvNeXtV2-tiny @224	0.15824
ConvNeXtV2-nano @128	0.15311
EfficientViT-b0 @128	0.13706
ViT-tiny @128	0.13654
MobileNetV4-small @128	0.11242

nano @224 matches tiny @224 at half the cost; chosen as the image expert.

COMBINER

Rank-Average Stack

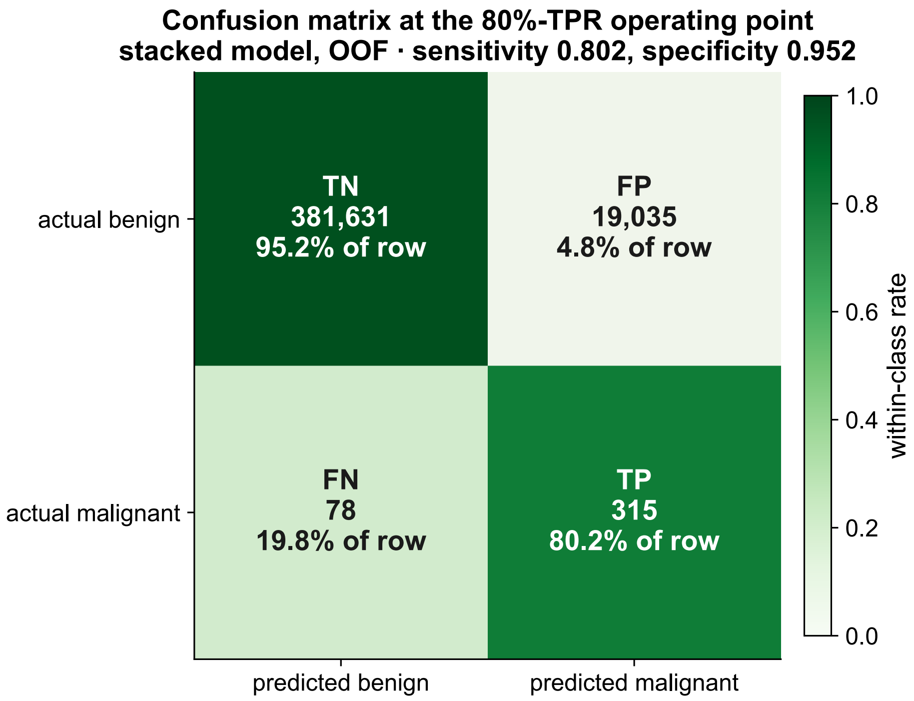
$$\hat{s}_i = \frac{1}{2}\left(r\left(p_i^{\text{tab}}\right) + r\left(p_i^{\text{img}}\right)\right), \qquad r(p_i) = \frac{\text{rank}(p_i)}{N} \in (0, 1]$$

- Each expert's scores are rank-transformed, then equally averaged.
- Tabular and image make **complementary errors**; the average sits above both on every fold.
- Tabular 0.16890 + image 0.15821 → stack **0.17376**.

WHY TRIVIAL WINS

A rank average has **zero learned parameters**, so it cannot overfit the 393 positives. A learned meta-stacker reaches only 0.17108; a learned per-lesion gate falls to 0.15007 — below tabular alone.

High Ranking, Prevalence-Limited Precision



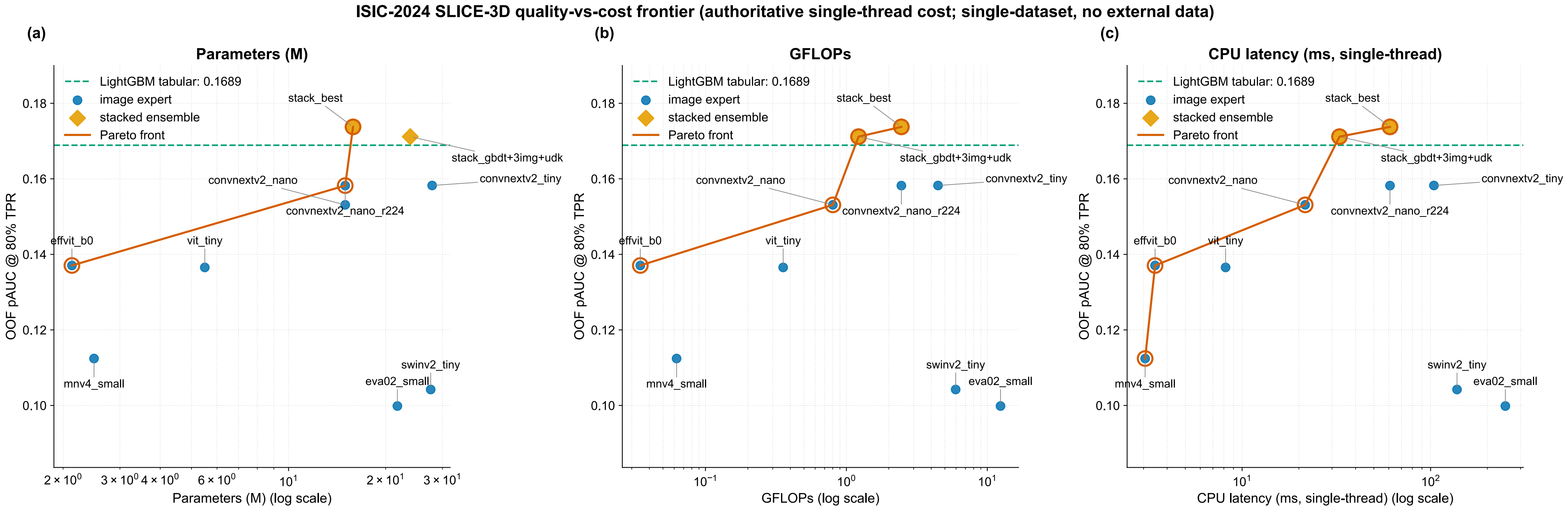
At the 80% TPR point: 315 caught, 78 missed, 19,035 false positives.

Stack, out-of-fold

pAUC @ 80% TPR (max 0.20)	**0.17376**
ROC-AUC	**0.9666**
Average precision	**0.094**
Sensitivity / specificity @ op. point	**0.802 / 0.952**

> ROC-AUC is high, but at **0.098% prevalence** precision is mathematically forced low — **which is why scoring is pAUC@80%TPR**, not precision or accuracy.

Quality vs. Cost: Three Axes



OOF pAUC against parameters, GFLOPs and single-thread CPU latency. The small-backbone stack owns the knee; heavy models are dominated.

Model	pAUC	Params (M)	GFLOPs	CPU (ms)	Pareto
Stack (rank-avg: GBDT + nano@224)	**0.17376**	**15.84**	**2.455**	**60.91**	**Yes**
Tabular GBDT (LightGBM + CatBoost)	0.16890	0.86	0.000	0.02	Yes
EVA-02-small @336 (heavy ViT, dominated)	0.09984	21.74	12.409	249.18	—

Added Complexity Did Not Pay

Ablations that did not improve OOF pAUC (leak-verified)

Idea tried	Its pAUC	Beaten by	Honest finding
Learned per-lesion gate (MoE)	0.15007	0.16890 (tabular)	loses to tabular alone
Meta-LGBM stacker	0.17108	0.17376 (rank-avg)	loses to trivial rank-avg
+ PCA image embeddings into GBDT	< 0.1743	0.17376	embeddings hurt
Heavy backbones (swinv2 / eva02)	0.104 / 0.100	0.15821 (nano@224)	dominated, overfit @393 pos
EMA 0.999 (over-smoothing)	~0.146	0.15821 (EMA0.995)	stronger EMA hurts

Every complexity increment lost: learned gate, meta-stacker, PCA embeddings, heavy backbones, stronger EMA.

Table 3. Comparison to top Kaggle solutions.

Solution	Ext.	Syn.	pAUC
1st — Novoselskiy (EVA-02 + GBDT)	Yes	Yes	0.17264
2nd — uchiyama33	Yes	Yes	—
3rd — kyohei-123	Yes	Yes	—
Ours (single-dataset)	**No**	**No**	**CV 0.17376**

Champions used banned external dermoscopy and ~30k synthetic positives; their own ablation reports synthetic data added only +0.0007. Ours is OOF CV, not private LB.

The Rulebook, and What It Costs

Self-imposed constraints

- **No external data.** SLICE-3D only — no ISIC-2019/2020, no PAD-UFES, no external dermoscopy.
- **No synthetic lesions.** Fabricated pathology is a label-validity hole in a medical task.
- ImageNet-pretrained encoders allowed: no skin-cancer labels, so the no-external-data property holds.

Limitations

- Only 393 positives bound model capacity; heavy backbones overfit.
- OOF pAUC is 0.17376; allowing the usual public-to-private drop, we estimate private ≈ 0.16.
- Precision at the operating point is low by construction at this prevalence.

CONCLUSION

Efficient, Honest Choices Won

- A small, cheap stack reaches **0.17376** OOF pAUC on a leak-verified patient-grouped split.
- Patient-relative tabular features carry most of the score.
- A zero-parameter rank average beats every learned combiner.
- The result holds **without external or synthetic data**.

CLAIM

SOTA among single-dataset, no-external-data, no-synthetic solutions, reported on a quality-versus-cost frontier rather than a single leaderboard point.

At 393 positives, the efficient choices were not only cheaper — they were **more accurate**.